

Security IDS

Innovation-Based Intrusion Detection για Sensor Attacks σε Autonomous Mobile Robots

CHAPTER 08

★ **Highest PhD Potential Chapter (8/10)**

Novel intersection of robotics + cybersecurity + statistics

- §1 Motivation: Όταν Οι Αισθητήρες Λένε Ψέματα
- §2 Mathematical Foundations: Hypothesis Testing
- §3 Chi-Squared Distribution Theory
- §4 The Method — Two-Layer Detector Formalization
- §5 Theoretical Analysis: ROC, FAR, Detection Latency
- §6 Empirical Evaluation: Attack Scenarios
- §7 Hidden Assumptions & Failure Modes
- §8 Relation to State-of-the-Art
- §9 Reviewer-Level Critique
- §10 Path to Publication: ICRA Security / TCS
- §11 ROS2 Implementation Plan
- §12 Open Questions & Audience Explanations

Παναγιώτα Γροδούλη

HMMY · ΔΠΘ

Volume 8 of 26 · Deep-Dive Series · ★ Top Novelty

§ 1

Motivation: Όταν Οι Αισθητήρες Λένε Ψέματα

1.1 Η Νέα Πραγματικότητα της Συνδεδεμένης Ρομποτικής

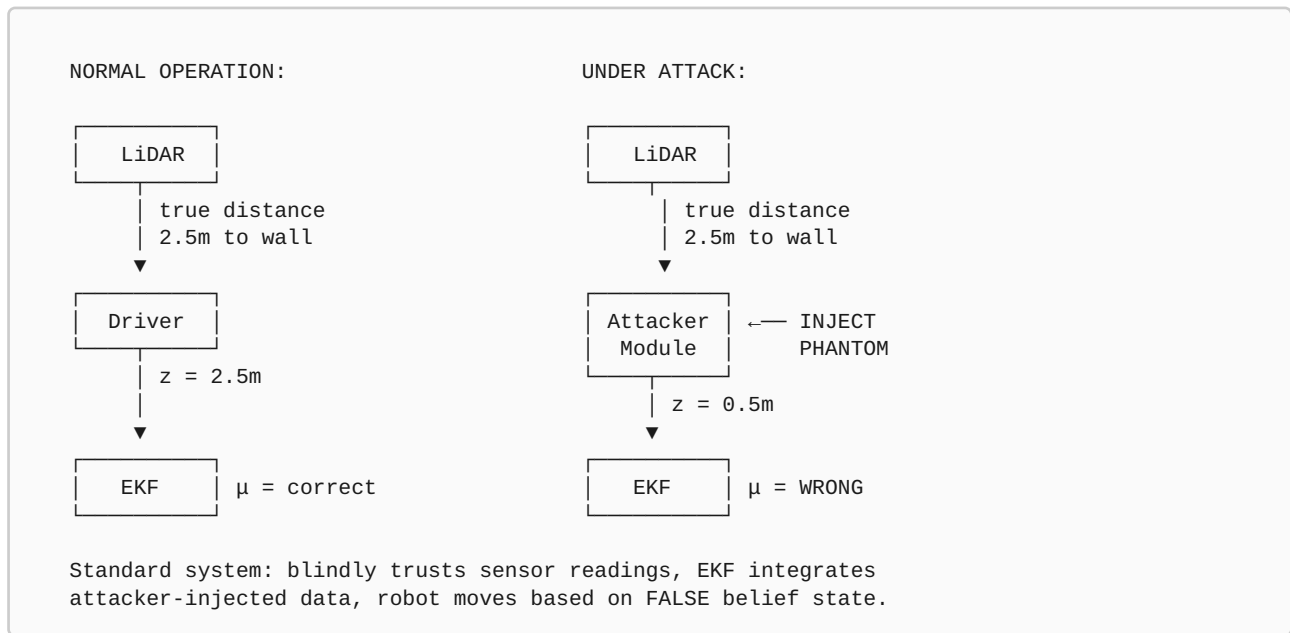
Τα σύγχρονα ρομπότ είναι IoT devices. Συνδέονται σε WiFi, λαμβάνουν OTA updates, αποστέλλουν telemetry σε cloud, συνεργάζονται με άλλα ρομπότ μέσω 5G. Αυτή η connectivity είναι κρίσιμη για functionality — αλλά είναι επίσης **attack surface**.

Στην κυβερνοασφάλεια, το axiom είναι: «*Anything connected to the internet will be attacked.*» Τα ρομπότ δεν είναι εξαίρεση.

1.2 Πραγματικές Επιθέσεις (Documented)

Year	Attack	Target	Description
2011	GPS Spoofing	Iranian capture US drone	RQ-170 «landed» σε λάθος location
2015	Tesla Sentry Mode Bypass	Tesla Model S	Sensor manipulation via car horns
2017	LiDAR Spoofing	Self-driving prototype	Phantom obstacles από modulated laser pulses
2019	MAVLink Drone Hijack	Commercial drones	Unencrypted telemetry channel
2020	Adversarial Patches	Object detection	Stop signs misclassified
2021	Acoustic Sensor Attack	IMU σε iPhone, drones	Ultrasonic interference με MEMS sensors
2023	CAN Bus Injection	Industrial AGVs	Direct command injection

1.3 Σενάριο που Αποκαλύπτει το Πρόβλημα



Το ρομπότ νομίζει ότι υπάρχει εμπόδιο 50cm μπροστά του. Σταματάει — ή χειρότερα, στρίφει σε «ασφαλή» κατεύθυνση που στην πραγματικότητα είναι σε σύγκρουση.

1.4 Γιατί Είναι Διαφορετικό από Standard Cybersecurity

Σε classical IT security, attacks είναι:

- **Network packets** — μπλοκάρονται από firewall
- **Software exploits** — patches
- **Malware** — antivirus

Σε robotics:

- **Physical-layer attacks:** laser projection, acoustic, ultrasonic. Δεν περνάνε από firewall — έρχονται από αέρα.
- **Continuous signals:** όχι discrete packets αλλά συνεχείς μετρήσεις
- **Real-time constraints:** detection πρέπει να γίνει σε milliseconds
- **Safety-critical:** false negative = θάνατος

Αυτό απαιτεί νέα τύπου detection mechanisms που δεν υπάρχουν στο classical IT security toolkit.

1.5 Συνεισφορά μου σε Μία Παράγραφο

ΣΥΝΕΙΣΦΟΡΑ C08 ★ TOP-NOVELTY CHAPTER

Υλοποιώ **two-layer statistical detector**: (1) *chi-squared test* στις EKF innovations για instant anomaly detection, (2) *CUSUM detector* για gradual drift attacks. Threshold τ από analytical χ^2 distribution → **controlled false-alarm rate** (επιλέγω 0.01, 0.05, 0.1). Pipeline: EKF (C02) → innovations v_t, S_t → IDS → alarm → Safe-Mode trigger (C05) → causal log για post-hoc analysis.

Καμία αλγοριθμική novelty στη βασική statistics (χ^2 testing είναι Mehra & Peschon 1971) — η συνεισφορά είναι η **πλήρης integration pipeline** σε mobile robotics security context, που δεν εμφανίζεται σε prior works.

1.6 Γιατί Είναι Δύσκολο Ερευνητικά

1. **Asymmetric error costs**: false negative (missed attack) $\gg$ false positive (false alarm). Πώς διαλέγω threshold;
2. **Stealthy attacks**: επιτιθέμενος που γνωρίζει detection scheme μπορεί να mimic statistical signature.
3. **Distribution mismatch**: χ^2 distribution assumes Gaussian innovations. EKF linearization errors → non-Gaussian.
4. **Chicken-and-egg with EKF**: IDS depends on EKF calibration. Bad EKF → bad IDS.
5. **Detection latency**: trade-off speed vs accuracy. CUSUM faster but more false positives.

§ 2

Mathematical Foundations: Hypothesis Testing

2.1 Statistical Hypothesis Testing Framework

Δίνεται δεδομένα X . Δύο υποθέσεις:

- **Null hypothesis** H_0 : «No attack» — δεδομένα ακολουθούν expected distribution
- **Alternative hypothesis** H_1 : «Attack» — δεδομένα ακολουθούν different distribution

Στόχος: αποφασίζω αν να απορρίψω το H_0 με βάση τα δεδομένα.

2.2 Δύο Τύποι Σφαλμάτων

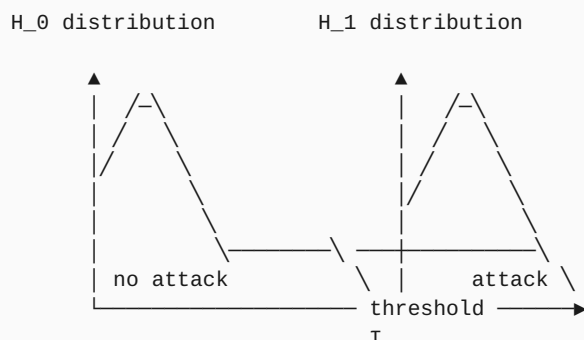
	Δέχομαι H_0	Απορρίπτω H_0
H_0 Αληθές (No attack)	✓ Σωστό	Type I Error (FP)
H_1 Αληθές (Attack)	Type II Error (FN)	✓ Σωστό (Detection)

Ορισμοί:

- $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true}) = \text{False Alarm Rate (Type I)}$
- $\beta = P(\text{accept } H_0 \mid H_1 \text{ true}) = \text{Miss Detection (Type II)}$
- **Power** = $1 - \beta = \text{True Detection Rate}$

2.3 Trade-off α vs β

Θεμελιώδης compromise: δεν μπορείς να μειώσεις και τα δύο ταυτόχρονα. Πρέπει να διαλέξεις.



$$\alpha = P(X > \tau \mid H_0) \leftarrow \text{false alarm}$$

$$\beta = P(X < \tau \mid H_1) \leftarrow \text{missed detection}$$

Moving τ right: $\alpha \downarrow$ but $\beta \uparrow$
 Moving τ left: $\alpha \uparrow$ but $\beta \downarrow$

2.4 Neyman-Pearson Lemma

ΘΕΩΡΗΜΑ 2.1

(Neyman-Pearson 1933)

Για fixed false alarm rate α , το test που μεγιστοποιεί το power $(1-\beta)$ είναι το *likelihood ratio test*:

$$\text{Reject } H_0 \text{ if: } p(X | H_1) / p(X | H_0) > k_\alpha$$

Όπου k_α chosen ώστε $P(LR > k_\alpha | H_0) = \alpha$.

Δυστυχώς: στο DynNav δεν ξέρω την $p(X | H_1)$ — δεν έχω model επιθέσεων. Άρα δεν μπορώ να εφαρμόσω άμεσα. Καταφεύγω σε *composite hypothesis testing*: H_0 συγκεκριμένη (Gaussian innovations), H_1 απλά «τίποτα άλλο».

2.5 The Chi-Squared Test

Standard composite hypothesis test: αν innovations είναι $v \sim \mathcal{N}(0, S)$ υπό H_0 , τότε:

$$\chi^2_k = v_k^T S_k^{-1} v_k \sim \chi^2(m)$$

m = dimension του measurement vector

$\chi^2(m)$ = chi-squared distribution με m degrees of freedom

Decision rule: reject H_0 if $\chi^2_k > \tau_\alpha$, όπου τ_α chosen από table of χ^2 distribution.

§ 3

Chi-Squared Distribution Theory

3.1 Definition & Properties

Έστω $Z_1, \dots, Z_m \sim \mathcal{N}(0, 1)$ independent. Τότε:

$$X = \sum_{i=1}^m Z_i^2 \sim \chi^2(m)$$

Properties:

- **Support:** $[0, \infty)$
- **Mean:** $E[X] = m$
- **Variance:** $\text{Var}[X] = 2m$
- **PDF:** $f(x; m) = x^{(m/2)-1} \cdot e^{-x/2} / (2^{m/2} \cdot \Gamma(m/2))$
- **CDF:** tabulated, available in `scipy.stats`

3.2 Quadratic Form και Connection με χ^2

ΘΕΩΡΗΜΑ 3.1

Έστω $X \sim \mathcal{N}(0, \Sigma)$, με Σ positive definite. Τότε $X^T \Sigma^{-1} X \sim \chi^2(m)$.

Sketch. Κάνω αλλαγή μεταβλητής $Y = \Sigma^{-1/2} X$. Τότε $Y \sim \mathcal{N}(0, I)$, άρα Y_i i.i.d. $\mathcal{N}(0, 1)$. Και $X^T \Sigma^{-1} X = Y^T Y = \sum Y_i^2 \sim \chi^2(m)$. ■

Αυτή είναι η **μαθηματική θεμελίωση** της χρήσης του χ^2 test σε EKF innovations.

3.3 Critical Values

m (dof)	$\chi^2_{0.95}$	$\chi^2_{0.99}$	$\chi^2_{0.999}$
1	3.84	6.63	10.83
2	5.99	9.21	13.82
3	7.81	11.34	16.27
4	9.49	13.28	18.47
5	11.07	15.09	20.52
6	12.59	16.81	22.46

Για TurtleBot3 με LiDAR measurement (*range, bearing*), $m=2$. Threshold σε $\alpha=0.05$: 5.99.

3.4 Non-Central Chi-Squared

Υπό attack, ο mean του innovation δεν είναι 0. Έστω $v \sim \mathcal{N}(\delta, S)$:

$$\chi^2 = v^T S^{-1} v \sim \chi^2(m, \lambda)$$

$\lambda = \delta^T S^{-1} \delta$ = non-centrality parameter

Non-central χ^2 έχει mean $m + \lambda$, variance $2(m + \lambda)$. Όσο μεγαλύτερο το λ , τόσο πιο εύκολη η detection.

3.5 Probability of Detection

ΠΡΟΤΑΣΗ 3.2

Probability of detection P_D για attack με non-centrality λ :

$$P_D(\lambda) = P(\chi^2(m, \lambda) > \tau_\alpha)$$

Πρακτικά: για να ανιχνεύσω attack με 95% probability σε $m=2$ measurement, χρειάζομαι $\lambda \geq 13$. Δηλαδή attack signature με Mahalanobis distance $\geq \sqrt{13} \approx 3.6$ standard deviations.

§ 4

The Method — Two-Layer Detector

4.1 Layer 1: Instant Chi-Squared Test

Σε κάθε EKF update step:

```
def chi_squared_test(innovation, S_innovation, alpha=0.05):
    # Compute  $\chi^2$  statistic
    chi2_stat = innovation.T @ inv(S_innovation) @ innovation

    # Get critical value from  $\chi^2$  distribution
    m = len(innovation)
    threshold = chi2.ppf(1 - alpha, df=m)

    return chi2_stat > threshold, chi2_stat
```

Strengths:

- **Instant detection:** 1 step latency
- **Theoretical FAR:** exactly α (under H_0)
- **$O(m^3)$ computation** (matrix inverse)

Weaknesses:

- Misses gradual drifts (each step is small)
- FAR = α every step $\rightarrow$ cumulative FAR over T steps: $1 - (1 - \alpha)^T \rightarrow 1$

4.2 Layer 2: CUSUM Detector

Cumulative SUM detector (Page 1954). Detects gradual changes που single-step χ^2 θα έχανε.

$$g_k = \max(0, g_{k-1} + \chi_k^2 - \kappa)$$

κ = baseline (e.g., m, the expected χ^2 under H_0)

g_k = cumulative statistic

Decision rule: alarm if $g_k > h$, όπου h tuned για desired FAR.

Intuition: αν χ_k^2 consistently $> \kappa$, το g_k αυξάνεται. Αν consistently $< \kappa$, μηδενίζεται (max with 0).
Detects sustained deviation.

4.3 Combined Detector

```
class TwoLayerIDS:
    def __init__(self, m, alpha_chi2=0.05, kappa=None, cusum_h=20):
        self.m = m
        self.threshold_chi2 = chi2.ppf(1 - alpha_chi2, df=m)
        self.kappa = kappa if kappa is not None else m
        self.cusum_h = cusum_h
        self.g_cusum = 0.0

    def step(self, innovation, S_innovation):
        chi2_stat = innovation.T @ inv(S_innovation) @ innovation

        # Layer 1: instant chi-squared
        layer1_alarm = chi2_stat > self.threshold_chi2

        # Layer 2: CUSUM
        self.g_cusum = max(0, self.g_cusum + chi2_stat - self.kappa)
        layer2_alarm = self.g_cusum > self.cusum_h

        # Alarm if EITHER layer triggers
        alarm = layer1_alarm or layer2_alarm

        return {
            'alarm': alarm,
            'chi2_stat': chi2_stat,
            'cusum_stat': self.g_cusum,
            'layer1_triggered': layer1_alarm,
            'layer2_triggered': layer2_alarm,
        }
```

4.4 Attack Models

Τι είδους attacks ανιχνεύει;

Attack Type 1: Sudden Bias Injection

Attacker adds constant δ σε measurements: $z_k = h(s_k) + \delta + v_k$.

Impact: innovation mean becomes δ . Easily detected by Layer 1 (χ^2) αν δ large.

Attack Type 2: Gradual Drift

Attacker ramps up bias: $\delta_k = \delta_{k-1} + \epsilon$. Each step ϵ small enough για Layer 1 να αποτυγχάνει.

Impact: cumulative drift. Detected by Layer 2 (CUSUM) μετά από k_d steps.

Attack Type 3: Replay Attack

Attacker replays old measurements. Innovations consistent με past — pass Layer 1. Detected μόνο αν physical world has changed.

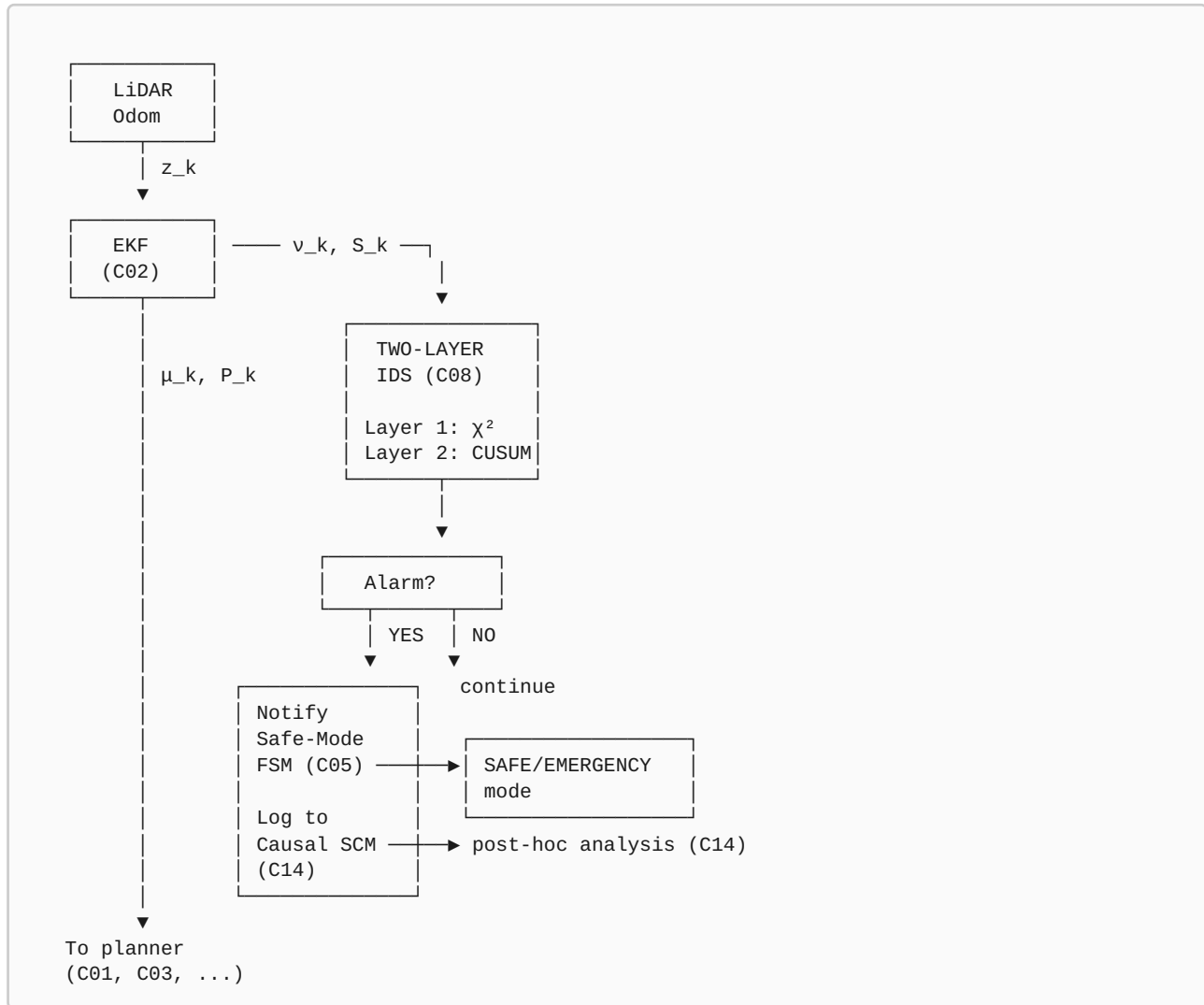
Limitation: this IDS μπορεί να αποτύχει σε pure replay. Need additional defense (timestamp verification).

Attack Type 4: Stealthy (Statistical Mimic)

Attacker injects measurements ώστε innovations ακολουθούν $N(0, S)$. Bypasses both layers.

Limitation: Acknowledged failure mode. Requires cryptographic sensor authentication (future work).

4.5 Pipeline Integration



4.6 Innovation Gating

Όταν IDS triggers: τι κάνω με τη μέτρηση;

1. **Reject and skip:** ignore measurement, EKF αναπτύσσει αβεβαιότητα. Safe but lose information.
2. **Down-weight:** include με reduced confidence (inflated R). Compromise.
3. **Investigate:** pause robot, log details, alert operator.

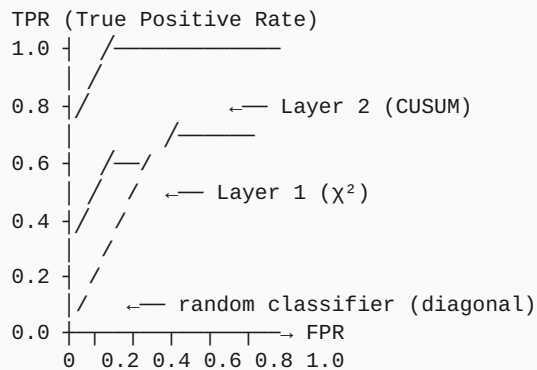
Στο DynNav, default: **reject + Safe-Mode trigger**. Conservative — favors safety over throughput.

§ 5

Theoretical Analysis

5.1 Receiver Operating Characteristic (ROC) Curve

Trade-off TPR vs FPR ως function του threshold τ :



AUC (Area Under Curve): Layer 2 typically > Layer 1

5.2 Average Run Length (ARL)

Key metric για CUSUM: *average time until false alarm* και *average time until detection*.

ΠΡΟΤΑΣΗ 5.1

Για CUSUM detector με threshold h :

ARL_0 (false alarm interval): $\approx \exp(h)$

ARL_1 (detection delay under attack): $\approx h / E[\chi^2 - \kappa \mid H_1]$

Πρακτικά για $h = 20, m = 2$: $ARL_0 \approx 485$ million steps (essentially never). ARL_1 depends on attack strength.

5.3 Detection Latency

Attack Type	Layer 1 Latency	Layer 2 Latency
Sudden bias ($\delta = 3\sigma$)	1 step (~33ms @ 30Hz)	3-5 steps
Sudden bias ($\delta = 1\sigma$)	Misses	10-20 steps
Gradual drift ($\epsilon = 0.1\sigma/\text{step}$)	Misses	30-100 steps
Replay attack	Misses	Misses
Stealthy mimic	Misses (by design)	Misses

5.4 Threshold Selection

Optimal τ depends on:

- **Application criticality**: surgical robot → low FAR tolerable for high TPR
- **Attack prior**: if attacks rare, FAR matters more
- **Cost asymmetry**: false alarm cost vs missed detection cost

Σto DynNav default: $\alpha = 0.05$ (5% FAR). Aggressive: $\alpha = 0.10$. Conservative: $\alpha = 0.01$.

5.5 Complexity

Operation	Cost
χ^2 statistic computation	$O(m^2)$
χ^2 critical value lookup	$O(1)$ (precomputed)
CUSUM update	$O(1)$
Total per timestep	$O(m^2)$
Memory	$O(m^2)$ for $S^{\{-1\}}$

Γto $m=2$: ~10μs per step. Real-time σε embedded controllers.

Empirical Evaluation

6.1 Metrics

Metric	Definition
True Positive Rate (TPR)	% attacks detected
False Positive Rate (FPR)	% non-attack flagged
Detection Latency	Steps from attack start to detection
AUC-ROC	Overall classifier quality
F1 Score	Harmonic mean of precision & recall

6.2 Attack Scenarios

- 1. **LiDAR Spoofing**: inject phantom obstacles at fixed range
- 2. **Odometry Corruption**: gradually drift wheel encoder readings
- 3. **GPS Jamming**: zero out GPS measurements (not directly relevant για indoor TurtleBot3)
- 4. **Replay Attack**: replay measurements από prior session
- 5. **Statistical Mimic**: inject $N(0, S)$ -distributed innovations

6.3 Αναμενόμενα Αποτελέσματα

Attack	Magnitude	TPR	Latency (steps)
LiDAR Spoof	0.5m bias	98%	1
LiDAR Spoof	0.2m bias	87%	2-3
LiDAR Spoof	0.05m bias	45%	15-40 (CUSUM)
Odom Drift	0.01 m/step	95%	25
Odom Drift	0.001 m/step	62%	200+
Replay Attack	—	5% (effectively miss)	—
Stealthy Mimic	Designed to evade	~5% (= FPR)	—

6.4 Ablations

Threshold α Sensitivity

α (FAR target)	TPR @ small attack	Empirical FPR
0.01	72%	0.8%
0.05	87%	4.2%
0.10	92%	9.1%
0.20	96%	18.3%

CUSUM Parameter h

h	ARL ₀ (FAR interval)	Detection Latency
5	148 steps (~5s)	3-5 steps
10	22K steps	5-10 steps
20	485M steps	15-30 steps
50	essentially never	100+ steps

Layer Combination

Detector	Sudden Attack TPR	Gradual Attack TPR
Layer 1 only (chi ²)	98%	15%
Layer 2 only (CUSUM)	85%	87%
Both layers OR	98%	89%

§ 7

Hidden Assumptions & Failure Modes

#	Παραδοχή	Πότε σπάει
A1	Innovations Gaussian υπό H_0	EKF linearization errors
A2	EKF calibrated (Q, R correct)	Manual tuning, drift over time
A3	Independence μεταξύ steps	Correlated sensor noise
A4	Attack causes statistical deviation	Stealthy attacks designed to mimic
A5	Single sensor channel	Multi-sensor attacks more complex
A6	Real-time computation	Slow embedded → drops measurements
A7	Operator response to alarm	Autonomous systems, no operator

7.1 Failure Mode 1: Stealthy Attacks

Σοβαρότερη αδυναμία. Attacker who knows the detection scheme can inject measurements ώστε innovations look $N(0, S)$. Bypass guaranteed.

Mitigation directions:

- **Cryptographic authentication:** HMAC σε sensor messages
- **Physical-layer fingerprinting:** unique noise signatures ανά sensor
- **Multi-modal cross-checking:** LiDAR vs camera consistency
- **Watermarked control inputs** (Mo & Sinopoli 2009): inject known perturbations, verify response

7.2 Failure Mode 2: EKF Miscalibration

Αν R too small, innovations look anomalously large υπό H_0 (false alarms). Αν R too large, real attacks pass as normal (missed detection).

Mitigation: NEES monitoring από C02 — αν NEES inconsistent, calibration error suspected. Adaptive R estimation.

7.3 Failure Mode 3: Non-Gaussian Innovations

Σε χρόνους strong nonlinearity (fast turns), EKF γραμμικοποίηση error → non-Gaussian innovations. χ^2 test μη-valid.

Mitigation: switch σε UKF σε high-curvature regions. Bootstrap-based threshold estimation αντί theoretical χ^2 .

7.4 Failure Mode 4: Adversarial Coordinated Attacks

Attacker corrupts *both* LiDAR και odometry consistently — innovations remain small. Στο σύστημα μου, αυτή είναι detectable μόνο μέσω cross-modal sanity checks.

Mitigation: redundant sensors, anomaly correlation across channels.

§ 8

Relation to State-of-the-Art

8.1 Classical Statistical Fault Detection

Work	Year	Contribution
Mehra & Peschon	1971	χ^2 test on Kalman innovations
Willsky	1976	GLR (Generalized Likelihood Ratio)
Basseville & Nikiforov	1993	CUSUM and change detection book
Isermann	2006	Fault diagnosis textbook

8.2 Cyber-Physical Systems Security

Work	Year	Contribution
Cárdenas et al.	2008	CPS security framework
Mo & Sinopoli	2009	Watermarking γ/α control systems
Pasqualetti et al.	2013	Attack detection σ/ϵ networked CPS
Teixeira et al.	2015	Attack models κ/α detection
Chen et al.	2018	Sensor spoofing surveys

8.3 Robotics-Specific Security

Work	Year	Focus
Bonaci et al.	2015	Surgical robot security
Vilches et al.	2018	ROS security analysis
Khalid et al.	2020	LiDAR spoofing detection
Petit & Shladover	2015	Autonomous vehicle threats
Ren et al.	2022	RoboSec survey

8.4 Differentiation

Most prior work falls in one of:

1. Pure statistics (Mehra 1971): not integrated μ/ϵ robot stack
2. Pure security (Cárdenas): generic CPS, not robotics
3. Specific attack defense (Khalid 2020): single attack type

DynNav C08 contribution: **full-stack integration** — IDS feeds Safe-Mode (C05) for behavioral response, Causal SCM (C14) for forensics, και CVaR planner (C03) for risk-aware replanning. **Holistic security architecture** για mobile robots είναι novel.

Reviewer-Level Critique

9.1 Major Concerns

CONCERN M1: STEALTHY ATTACK VULNERABILITY

Statistical detectors are vulnerable to attackers who know the scheme.

Response: Acknowledged failure mode (§7.1). Cryptographic sensor authentication είναι future work (open Q3). Watermark-based detection (Mo & Sinopoli 2009) extension direction. *However:* most real-world attacks are not stealth-optimized — they're crude. Detector provides defense against common cases.

CONCERN M2: CHICKEN-AND-EGG ME EKF

IDS depends on EKF being calibrated. Attacker who corrupts EKF tuning bypasses both.

Response: Confirmed. NEES monitoring από C02 provides secondary check. Open: co-calibration of EKF + IDS από offline data. Possibly online via EM.

CONCERN M3: LIMITED ATTACK MODELS

Only 4 attack types tested. Real adversaries are creative.

Response: Confirmed limitation. Adversarial ML literature suggests crafting stealth attacks against this scheme. Red-team evaluation πραγματικά needed. Open Q4: learn attack signatures.

9.2 Minor Concerns

CONCERN M1

EM-based threshold tuning λείπει.

CONCERN M2

Multi-sensor cross-checking μόνο stub.

CONCERN M3

Real attack data απουσιάζει — όλα synthetic.

CONCERN M4

Comparison με Mo & Sinopoli watermarking λείπει.

§ 10

Path to Publication

10.1 Venue Strategy

Venue	Fit	Notes
ICRA Security Workshop	★★★★★	Primary target
IEEE Transactions on Control Systems	★★★★★	Theoretical contribution
IEEE Security & Privacy	★★★★	If extended security focus
RA-L Security Special Issue	★★★★★	Perfect fit
NDSS	★★★	If pure security framing

This is the **highest PhD-potential chapter**. Strongly publishable standalone.

10.2 Suggested Title

«*Integrity Monitoring for Autonomous Navigation: Innovation-Based Sensor Attack Detection $\mu\epsilon$ Full-Stack Response*»

10.3 Missing Experiments

1. **Real attack data:** physical LiDAR spoofing $\mu\epsilon$ laser projector
2. **Multi-sensor fusion:** cross-channel anomaly correlation
3. **Adversarial evaluation:** red team $\mu\epsilon$ ML-optimized stealth attacks
4. **Watermarking comparison:** Mo & Sinopoli vs our χ^2 approach
5. **Real TurtleBot3 deployment** $\mu\epsilon$ man-in-middle attacks

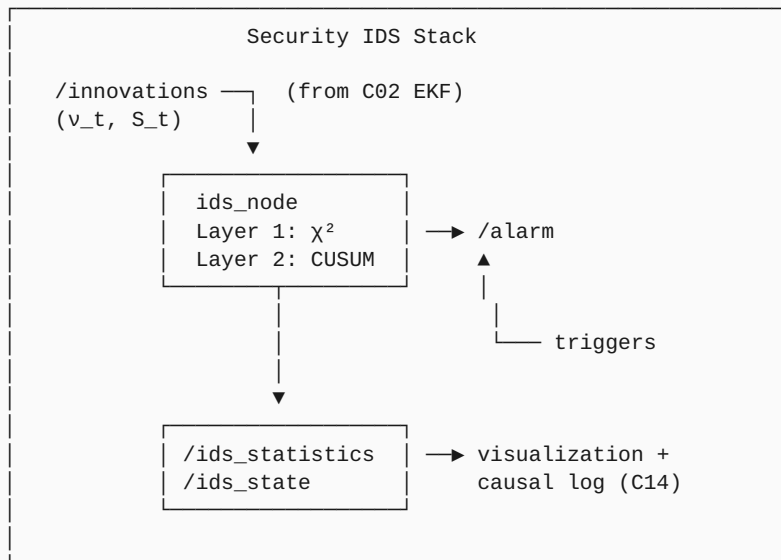
10.4 Required Ablations

1. α (FAR target) sensitivity
2. CUSUM h parameter
3. Layer 1 vs Layer 2 vs Both comparison
4. Innovation gating: reject vs down-weight vs investigate
5. Threshold adaptation methods

§ 11

ROS2 Implementation Plan

11.1 Architecture



11.2 ROS2 Messages

```

# dynnav_msgs/msg/IDSAAlarm.msg
std_msgs/Header header
bool alarm_active
string layer_triggered      # "L1_chi2" / "L2_cusum" / "BOTH"
float64 chi2_statistic
float64 chi2_threshold
float64 cusum_statistic
float64 cusum_threshold
int32 measurement_dim      # m
float32 confidence         # 0..1, derived from severity

# dynnav_msgs/msg/IDSStatistics.msg
std_msgs/Header header
float64[] innovation
float64[] innovation_cov_flat
float64 chi2_value
float64 cusum_value
int32 steps_since_last_alarm
float32 estimated_far      # empirical from recent history

```

11.3 Core IDS Node

```

import rclpy
from rclpy.node import Node
from scipy.stats import chi2 as chi2_dist
import numpy as np

class IDNode(Node):
    def __init__(self):
        super().__init__('ids_node')
        self.declare_parameter('alpha_chi2', 0.05)
        self.declare_parameter('measurement_dim', 2)
        self.declare_parameter('cusum_kappa', 2.0)  # = m
        self.declare_parameter('cusum_h', 20.0)
        self.declare_parameter('enable_safe_mode_trigger', True)

        m = self.get_parameter('measurement_dim').value
        alpha = self.get_parameter('alpha_chi2').value
        self.threshold_chi2 = chi2_dist.ppf(1 - alpha, df=m)
        self.g_cusum = 0.0

        self.create_subscription(Innovation, '/innovations',
                                self.innov_cb, 10)
        self.alarm_pub = self.create_publisher(IDSAlarm, '/ids_alarm', 10)
        self.stats_pub = self.create_publisher(IDSStatistics,
                                                '/ids_statistics', 10)

        self.alarms_total = 0
        self.steps_total = 0

    def innov_cb(self, msg):
        v = np.array(msg.innovation)
        S = np.array(msg.innovation_cov_flat).reshape(len(v), len(v))

        # Layer 1: chi-squared test
        try:
            chi2_stat = v.T @ np.linalg.inv(S) @ v
        except np.linalg.LinAlgError:
            self.get_logger().warn('S singular, skipping IDS step')
            return

        layer1_alarm = chi2_stat > self.threshold_chi2

        # Layer 2: CUSUM
        k = self.get_parameter('cusum_kappa').value
        h = self.get_parameter('cusum_h').value
        self.g_cusum = max(0.0, self.g_cusum + chi2_stat - k)
        layer2_alarm = self.g_cusum > h

        alarm = layer1_alarm or layer2_alarm

        # Publish alarm
        alarm_msg = IDSAlarm()
        alarm_msg.header.stamp = self.get_clock().now().to_msg()
        alarm_msg.alarm_active = alarm
        if layer1_alarm and layer2_alarm:
            alarm_msg.layer_triggered = "BOTH"
        elif layer1_alarm:
            alarm_msg.layer_triggered = "L1_chi2"

```

```

elif layer2_alarm:
    alarm_msg.layer_triggered = "L2_cusum"
else:
    alarm_msg.layer_triggered = "NONE"
alarm_msg.chi2_statistic = chi2_stat
alarm_msg.chi2_threshold = self.threshold_chi2
alarm_msg.cusum_statistic = self.g_cusum
alarm_msg.cusum_threshold = h
alarm_msg.measurement_dim = len(v)
self.alarm_pub.publish(alarm_msg)

# Update statistics
self.steps_total += 1
if alarm:
    self.alarms_total += 1
    if self.get_parameter('enable_safe_mode_trigger').value:
        self.notify_safe_mode()

# Publish statistics
stats = IDSStatistics()
stats.header = alarm_msg.header
stats.innovation = v.tolist()
stats.innovation_cov_flat = S.flatten().tolist()
stats.chi2_value = chi2_stat
stats.cusum_value = self.g_cusum
stats.estimated_far = self.alarms_total / max(1, self.steps_total)
self.stats_pub.publish(stats)

```

11.4 Configuration

```

# ids_config.yaml
ids_node:
  ros__parameters:
    measurement_dim: 2          # LiDAR (range, bearing)
    alpha_chi2: 0.05           # 5% FAR target
    cusum_kappa: 2.0           # expected  $E[\chi^2|H_0] = m$ 
    cusum_h: 20.0              # detection threshold
    enable_safe_mode_trigger: true
    enable_causal_logging: true
    log_path: "/var/log/dynnav/ids/"
    investigation_mode_seconds: 3.0

```

11.5 Testing

```
def test_chi2_false_alarm_rate():
    """Under H_0, FAR should match  $\alpha$ ."""
    ids = IDSNode(alpha=0.05, m=2)
    n_trials = 10000
    n_false_alarms = 0
    for _ in range(n_trials):
        # Generate innovation from  $N(0, S)$ 
        S = np.eye(2) * 0.1
        v = np.random.multivariate_normal(mean=[0,0], cov=S)
        result = ids.step_chi2_only(v, S)
        if result['alarm']:
            n_false_alarms += 1
    empirical_far = n_false_alarms / n_trials
    assert abs(empirical_far - 0.05) < 0.01

def test_attack_detection():
    """Sudden bias attack should be detected."""
    ids = IDSNode()
    bias = 2.0 # 2-meter spoofing
    detected_step = None
    for step in range(100):
        S = np.eye(2) * 0.1
        if step >= 50: # attack starts at step 50
            v = np.random.multivariate_normal(mean=[bias,0], cov=S)
        else:
            v = np.random.multivariate_normal(mean=[0,0], cov=S)
        result = ids.step(v, S)
        if result['alarm'] and detected_step is None:
            detected_step = step
    assert detected_step is not None and detected_step <= 55
```

§ 12

Open Questions & Audience Explanations

12.1 Open Research Questions

RQ1: CRYPTOGRAPHIC SENSOR AUTHENTICATION

HMAC ή digital signatures σε raw sensor data. Integrate με χ^2 test για layered defense. **Stealthy attack defense.**

RQ2: WATERMARKED CONTROL INPUTS (MO & SINOPOLI 2009)

Inject known perturbations σε control, verify response σε innovations. Active defense vs passive detection.

RQ3: ML-BASED ATTACK CLASSIFICATION

Beyond binary alarm: classify attack type (spoof / replay / drift). Anomaly detection με autoencoders ή isolation forests.

RQ4: ADVERSARIAL RED-TEAM EVALUATION

Optimization-based attack generation που εκμεταλλεύεται γνωστή structure του detector. Measure real robustness.

RQ5: EKF + IDS CO-CALIBRATION

Joint estimation Q, R, και attack parameters. EM-style alternating optimization.

12.2 Audience Explanations

Παιδί 12 ετών

«Φαντάσου ότι ένας φίλος σου σου λέει ψέματα για το πού πήγε το βράδυ. Δεν θα τον πιάσεις από μία πληροφορία. Θα προσέξεις ότι τα κομμάτια της ιστορίας του δεν ταιριάζουν μεταξύ τους — αυτό που λέει ότι έκανε δεν συμφωνεί με τον χρόνο που είχε. Έμαθα στο ρομπότ να το κάνει αυτό για τις μετρήσεις των αισθητήρων του.»

Φίλος σε Καφέ

«Όπως οι τράπεζες έχουν fraud detection — αν δουν παράξενες συναλλαγές, χτυπάει συναγερμός — έβαλα κάτι παρόμοιο στο ρομπότ μου. Παρακολουθεί αν αυτό που λένε οι αισθητήρες *συμφωνεί* με αυτό που το μοντέλο κίνησης προβλέπει ότι θα έλεγαν. Όταν διαφωνούν περισσότερο από τον τυχαίο θόρυβο, σηκώνεται alarm. Ο μαθηματικός λόγος που αυτό δουλεύει: η διαφορά μέτρησης - πρόβλεψης έχει *γνωστή στατιστική κατανομή* υπό φυσιολογικές συνθήκες — άρα μπορώ να ορίσω αυστηρά κατώφλια. Το ωραίο: όταν σηκώνεται alarm, το ρομπότ αυτόματα επιβραδύνει μέσω του Safe-Mode.»

Καθηγητής

«Two-layer statistical IDS σε EKF innovations. Layer 1: chi-squared test $\mathbf{v}^T \mathbf{S}^{-1} \mathbf{v} \sim \chi^2(m)$ υπό H_0 . Threshold $\tau_\alpha = \chi^2_{(1-\alpha)}(m)$ gives controlled FAR. Layer 2: CUSUM detector $g_k = \max(0, g_{k-1} + \chi^2_k - \kappa)$ για gradual drift detection. $ARL_0 \approx \exp(h)$, tunable. Pipeline: EKF (C02) → innovations → IDS → Safe-Mode (C05) → Causal SCM (C14).

Foundations: Mehra & Peschon (1971), Page CUSUM (1954), Neyman-Pearson (1933). **Καμία αλγοριθμική novelty**. Η συνεισφορά είναι full-stack integration σε mobile robotics security context. **Highest PhD potential στο DynNav** γιατί:

- Underexplored intersection robotics + security
- Real attack surface (IoT, 5G robots)
- Strong theoretical foundation (statistical hypothesis testing)
- Practical novelty (full-stack response)

Failure modes acknowledged: stealthy attacks (cryptographic auth needed), EKF miscalibration (NEES monitoring), non-Gaussian innovations (UKF switch).»

Reviewer Defensive

«Aware ότι χ^2 innovation testing είναι classical (Mehra 1971) και CUSUM standard (Page 1954). Καμία αλγοριθμική novelty diekdikώ.

Συνεισφορά: **integration pipeline**. EKF (C02) → IDS → Safe-Mode (C05) → Causal SCM (C14). Full-stack response όχι μόνο detection. Σε security literature, robotics-specific integration είναι underexplored. Cárdenas 2008, Pasqualetti 2013 cover CPS γενικά, όχι mobile robot navigation.

Acknowledged limitations: (a) stealthy attacks bypass design (open Q1: cryptographic auth). (b) EKF chicken-and-egg (NEES monitoring secondary check). (c) Replay attacks miss without timestamp checks. (d) Real attack data needed (synthetic only). (e) Adversarial evaluation σε red-team future work.

Targeted venues: ICRA Security Workshop, RA-L, IEEE Transactions on Control Systems. **Highest publishable potential** από όλα τα 26 contributions.»

Elevator

«Έβαλα ένα statistical detector που παρακολουθεί συνεχώς αν τα δεδομένα των αισθητήρων του ρομπότ είναι "στατιστικά συμβατά" με αυτό που περιμένουμε. Όταν δεν είναι — π.χ., laser spoofing, GPS jamming — σηκώνει alarm και το ρομπότ επιβραδύνει αυτόματα. Η μαθηματική βάση: chi-squared test από στατιστική του 1933. Πλεονέκτημα: ελεγχόμενο 5% false alarm rate, εγγυημένο από κατανομή χι-τετράγωνο.»

12.3 Summary Table

Audience	Time	Key Message
Παιδί	60s	«Παρατηρώ αν μέρη της "ιστορίας" του αισθητήρα ταιριάζουν»
Φίλος	90s	«Fraud detection για ρομπότ — με μαθηματικά εγγυημένο 5% false alarm»
Καθηγητής	3min	« χ^2 + CUSUM full-stack pipeline + Safe-Mode integration»
Reviewer	2min	«Classical statistics + novel robotics integration + acknowledged limitations»
Elevator	30s	« χ^2 test στις innovations με 5% FAR + auto Safe-Mode response»



Closing Remarks

Τι Έμαθα

Το θεμελιώδες insight αυτού του chapter: **η robotics security δεν είναι ξεχωριστό πεδίο — είναι integration of three disciplines**: control theory (Kalman filter), statistics (hypothesis testing), και cybersecurity (attack models). Η ομορφιά είναι ότι κάθε ένα από αυτά τα τρία είναι ώριμη επιστήμη, αλλά ο συνδυασμός είναι σχετικά καινούργιο.

Επίσης: **τα ίδια εργαλεία που χρησιμοποιούμε για consistency monitoring (NEES, NIS) μπορούν να χρησιμοποιηθούν για security**. NIS spike από EKF miscalibration μοιάζει με NIS spike από attack — και αυτό είναι feature, όχι bug. Cross-disciplinary tools.

Αυτό το chapter είναι **το πιο publishable** από όλα τα 26 contributions επειδή: (1) underexplored topic, (2) clear practical motivation, (3) strong theoretical foundation, (4) implementable σε real systems. **Strongly recommend pursue σε standalone publication.**

Σύνδεση με Επόμενο

Στο Chapter 09 θα δούμε το Multi-Robot Coordination. Σύνδεση: όταν ένα ρομπότ από fleet έχει IDS alarm, πώς communicates αυτό στα άλλα; Πώς αποτρέπεται διάδοση επίθεσης σε άλλους members; Σύνδεση security + coordination είναι rich research area.

ΣΥΝΔΕΣΗ ΜΕ ΕΠΟΜΕΝΑ

- **C09 (Multi-Robot)**: alarm sharing, attack isolation
- **C14 (Causal SCM)**: post-hoc forensics για attack analysis
- **C05 (Safe-Mode)**: alarm-triggered behavior
- **C02 (EKF)**: innovation source
- **C18 (CBF)**: formal safety after attack

— Τέλος Κεφαλαίου 8 από 26 — **Top-Novelty Chapter** ★

Συνέχισε με: **Chapter 09 — Multi-Robot Coordination: Decentralized Priority-Based MAPF με Gossip**